IEOR 221

Homework 9

Module 10 Exotic Options and Structured Products (November 5, 2025)

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1) Suppose that c_1 and p_1 are the prices of a European average price call and a European average price put with strike price K and maturity T, c_2 and p_2 are the prices of a European average strike call and European average strike put with maturity T, and c_3 and p_3 are the prices of a regular European call and a regular European put with strike price K and maturity T. Show that $c_1 + c_2 - c_3 = p_1 + p_2 - p_3$. (Hull 26.4)

2) What is the value of a derivative that pays off $100 in 6 months if an index is greater than 1,000 and zero otherwise? Assume that the current level of the index is 960, the riskfree rate is 8% per annum, the dividend yield on the index is 3% per annum, and the volatility of the index is 20%. (Hull 26.18)

3) Let S = 100, K = 80, r = 5%, q = 1%, sigma = 30%. Consider a perpetual American put option.
a) What is its fair value?
b) What is the value of its delta?
c) What is the value of its vega?